

CS 271 Computer Architecture and Assembly Language

Course Calendar* Summer 2024

*Weeks are shown Sunday-Sunday, all work is due on Sunday 11:59pm Pacific Time unless otherwise specified.
Schedule subject to change based on material pace.

Assignments opened that week are in BLACK. Assignments due that week are in RED.

Module	Topics
#1: June 23 rd – June 30 th Syllabus Quiz Module 1 Summary Exercises Project #0 Syllabus Quiz Module 1 Summary Exercises Project #0	• Introduction to Assembly Language • Internal Data Representation • Binary/Hexadecimal Arithmetic • Computer Architecture Fundamentals • IA-32 Basic Execution Environment
#2: June 30 th – July 7 th Module 2 Summary Exercises Project #1 Quiz #1 Module 2 Summary Exercises Quiz #1	• MASM Assembly ○ Introduction ○ How Instructions Work ○ Defining Data (Variables) ○ Arithmetic Operations • Irvine Procedures and your First Program
#3: July 7 th – July 14 th Module 3 Summary Exercises Project #2 Module 3 Summary Exercises Project #1 Project #2	• MASM Assembly ○ Conditions and Control Structures ○ Repetition Structures • Using the Visual Studio Debugger
#4: July 14 th – July 21 st Module 4 Summary Exercises Project #3 Quiz #2 Module 4 Summary Exercises Project #3 Quiz #2	• MASM Assembly ○ Defining Constants ○ Data Validation • Endianness • IEEE 754 Floating Point Format

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Module	Topics
#5: July 21 st – July 28 th Module 5 Summary Exercises Project #4 Module 5 Summary Exercises Project #4	• The Runtime Stack • Program Design (Modularization) • MASM Assembly ○ Procedure Calls/Returns ○ Procedure Documentation ○ Passing Parameters to Procedures
#6: July 28 th – August 4 th Module 6 Summary Exercises Project #5 Quiz #3 Module 6 Summary Exercises Quiz #3	• Error Detecting Codes • MASM Assembly ○ More Parameter Passing (Stack) ○ Arrays & Array parameters
#7: August 4 th – August 11 th Module 7 Summary Exercises Project #6 Module 7 Summary Exercises Project #5	• MASM Assembly ○ Data-related operators ○ String Processing ○ Macros • Reverse Polish Notation (RPN) • IA-32 floating-point unit (FPU)
#8: August 11 th – August 18 th Module 8 Summary Exercises Quiz #4 Module 8 Summary Exercises Project #6 Quiz #4	• CISC vs. RISC Architectures • Parallelism